



A Machine Learning Approach for Crime Classification and Prediction

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Abstract

Crime prediction has become an important challenge in modern cities wanting to enhance public safety and law enforcement. Analyzing crime patterns provides us a valuable insight on criminal behaviour and supports the efforts made for crime prevention. This study tackles the challenge of accurately classifying and forecasting criminal activities using machine learning methods. By examining spatial and time-based crime patterns, this study helps in identifying high-risk zones and predict future incidents based on key attributes such as location, time, seasonal crimes and other crime categories. Utilizing the publicly available Chicago crime dataset from Kaggle that includes a huge amount of city-level incident records, the study develops an integrated prediction framework combining Support Vector Machine (SVM) for crime classification and XGBoost for predictive analysis. To handle data imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is applied, improving the model's ability to learn minority crime classes effectively. Experimental evaluation shows that XGBoost achieves a high prediction accuracy of 97.3%, significantly outperforming SVM, which attains 53.2% accuracy. The results confirm that the combined use of SVM and XGBoost provides a reliable and scalable solution for crime prediction.

Keywords Crime classification · Crime prediction · SMOTE · Support vector machine · XGBoost

Introduction

Crime prediction and classification have become essential components of modern law enforcement, enabling better detection and prevention [1]. Understanding crime patterns requires analyzing diverse crime types and their frequency to understand the exact cause of their occurrence. As crime trends changes time to time due to various factors including socioeconomic, environmental and cultural differences, identifying high-risk areas known as the hotspots has become crucial for improving public safety [2]. However, the growing complexity and volume of crime data make manual analysis very difficult [3]. Even with better digital surveillance and record-keeping, accurate prediction of when and where the crime occurs still remains a major challenge.

The integration of machine learning with crime mapping has significantly advanced predictive policing by addressing spatial and temporal relationships within historical crime data [4]. These models assist in anticipating future incidents and categorizing criminal activities with greater accuracy [5]. Despite these improvements, prior studies have often

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overlooked key challenges such as data imbalance, lack of hybrid model integration and utilization of spatial-temporal attributes [6]. Many traditional models, including Decision Trees and Random Forests, provide good interpretability but tend to underperform in certain scenarios. Additionally, hybrid models that combine traditional machine learning techniques with deep learning approaches have demonstrated higher accuracy in crime prediction [7].

In this work, Support Vector Machine (SVM) and XGBoost were selected due to their complementary strengths. SVM works particularly well for high-dimensional data that can be separated linearly, which made it a good fit for classifying different crime types. On the other hand, XGBoost, being an ensemble gradient boosting model, offers strong predictive power and handles non-linear patterns effectively something crucial for time-based crime data. While models like Random Forest and LSTM were explored, Random Forest showed weaker precision when dealing with imbalanced datasets, and LSTM depended on detailed temporal features that our data lacked. Therefore, SVM and XGBoost were ultimately chosen for their balance of interpretability, efficiency, and accuracy.

Addressing these research gaps, this study developed a reliable framework that integrates Support Vector Machine (SVM) and XGBoost algorithms for crime classification and prediction. Through data preprocessing and model optimization the main contributions of this research include the following:

- developing a predictive framework using ML techniques.
- improving the model performance through parameter tuning.
- providing insights into different criminal behavior patterns for enhanced decision-making.

Together, they aim to enhance model reliability for real-world crime analysis.

Related Work

In previous years, researchers had explored enormous strategies for control of crime and a wide range of prediction algorithms has been discussed. However, effectiveness of algorithms tends to vary, depending on the quality of the data and features being used.

Previous studies on crime prediction can generally be grouped into three main categories. Spatial-temporal methods, such as STMEC proposed by Sun et al. [8], focus on integrating both spatial and temporal factors to detect crime hotspots and analyze evolving temporal trends. These approaches enable a deeper understanding of how crime

patterns change over time and across different locations. Deep learning models shown in the study Safat et al. [9], on the other hand use algorithms like Convolutional Neural Networks (CNNs), Deep Neural Networks (DNNs), and Long Short-Term Memory (LSTM) networks to capture complex, non-linear relationships among multiple variables. Additionally, hybrid machine learning approaches combine different algorithms often through ensemble frameworks to enhance the overall performance through improved accuracy and generalization across different datasets.

One notable study Jenga et al. [10] reviewed the use of machine learning algorithms in crime prediction and found that ANN(Artificial Neural Networks) are the most commonly applied techniques. The strength lies in their ability to model complex patterns within crime data, especially as when diverse and rich datasets are incorporated. Another study, Mandalapu et al. [11] analyzes more than 150 papers focused on deep learning and machine learning methods, categorizing algorithms such as random forests, decision trees and neural networks, evaluating the performance in identifying crime patterns and trends. Bhagat et al. [12] introduced a framework which combines machine learning with computer vision, showing how visual data interpretation can improve the accuracy and timeliness of crime forecasting. Similarly, Sun et al. [8] proposed the STMEC (SpatialTemporal Metapath Guided Explainable Crime Prediction) framework, building a heterogeneous graph of regions and attributes and were able to hold complex semantic relationships by factoring in both environmental and social data.

Another study, Joshi et al. [13] utilized crime data from Vancouver's open data catalog, implementing boosted decision tree algorithms and K-Nearest Neighbor (KNN). They achieved an accuracy of 39% and 44%, respectively. The model considers features like the date, time, location, and type of crime. Similarly, a systematic literature review, Artificial Intelligence & Crime Prediction [14] proctored 120 research papers and identified 64 distinct machine learning techniques applied to crime prediction. The review brings that supervised learning is the most widespread approach, lowering the importance of choosing algorithms tailored to specific crime categories. Further, Kamal Tanha [15] offers methodological taxonomy to categorize crime prediction techniques based on insights into how various data mining approaches perform. Lin et al. [16] focuses on grid based crime prediction using geographic features, employing a deep neural network (DNN) with tuning and achieved an accuracy of 82.79% and 81.45%.

Liu [17] uses the UCF101 dataset to classify behaviors such as vandalism, abuse and shooting into three different grades with achieving accuracy scores above 82% using AlphaPose and explored crime prediction using video data

and deep learning. Ogunde et al. [18] on a different approach applied ID3 decision tree algorithm to predict crimes and detect potential suspects based on previous data, reaching an accuracy of 72.73%. Tolan and Soliman [19] conducted experiments to forecast terrorism using classification algorithms like Naive Bayes, KNN, ID3 and SVM. Accuracy widely ranges from 26.01% to 75.41%, along with SVM performing the best under tenfold cross validation setup. Kavva et al. [20] also explores crime prediction using KNN and pruning techniques, reporting accuracy levels between 39% and 44%. Several studies has performed comparative analysis. Several studies, for example Babakura et al. [21] compared naïve Bayes and backpropagation algorithms for predicting crime categories. Traditionally, police departments has been relying on hotspot analysis Kouziokas [22], Umair et al. [23], where historical crime data is mapped to identify high risk areas and provide resources accordingly Chainey et al. [24].

Liberati et al. [25] examined crime from temporal and spatial perspective, applying the PRISMA methodology to systematically review the state-of-the-art crime prediction methods. Further, Walczak [26], Liu et al. [17] took an another angle using neural networks for assistance of police in decision-making. Although neural networks perfectly predicted a specific crime cluster only 16.4% of the time, it achieved an overall accuracy of 27.1%. Another study, Ayisheshim Alkaw and Kalyani Kadam [27] investigated different data mining and ensemble learning methods, analyzing factors like time, location, season, and day of the week how influence crime patterns. Similarly, McClendon and Meghanathan [28] applied supervised classification techniques to forecast crime types by analyzing labeled historical data. Meanwhile, Ippolito and Lozano [29] used models like neural networks and Naïve Bayes

for decision-making in fiscal audits in São Paulo. Finally, Falade et al. [30] highlight the crucial role that governments and law enforcement agencies play as gatekeepers of historical and real-time crime data, supporting for better data supervising and accessibility.

While these machine learning algorithms have significantly improved in the field of crime prediction, there's still room for improvement. It is also crucial to remember that not any single algorithm works best in all scenarios each dataset and its attributes can influence results, making performance evaluation an essential step in this process. Better results can be achieved through fine tuning of algorithm parameters, merging diverse data sources and carefully addressing of ethical concerns related to prediction policy.

A review of previous studies reveals several limitations. Many focus solely on either spatial-temporal analysis or categorical classification, without integrating both aspects into a single framework. Additionally, most rely on a single algorithm, which can reduce their ability to generalize across different datasets. Another common gap is the limited attention given to handling imbalanced data or comparing the performance of hybrid models. To address these issues, the present study proposes a two-stage hybrid approach—using SVM for classification and XGBoost for prediction—trained on real-world crime data from Chicago. The model also incorporates SMOTE to mitigate class imbalance and improve overall performance. Table 1 provides a comparative overview of several baseline crime prediction models from earlier studies alongside the proposed hybrid SVM+XGBoost approach. The results clearly highlights its improved effectiveness over traditional methods.

Proposed Framework

The framework shown in Fig. 1 represent a sequential hybrid model approach. Addressing class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) method is applied, to improve model's ability to recognize diminished crime categories. In the first stage, the SVM classifier categorizes crimes based on the given input features. The resulting labels are then fed into the XGBoost model, which predicts the likelihood of future incidents for each crime category. In this setup, SVM acts as the front-end classifier, while XGBoost serves as the temporal prediction model. By linking the two, the system allows classification results to guide prediction, leading to a more context-aware and informed learning process. The integration of SVM for classification and XGBoost for prediction improves the system's accuracy, ensuring reliable crime detection and prediction that aids law enforcement agencies in crime prevention and timely decision making ultimately contributing to reduced

Table 1 Comparison of performance of base models

Author & Year	Methodology	Dataset	Accuracy	Key findings
Liu [16]	Deep Learning (AlphaPose)	UCF101	82%	Video-based behavioural classification
Ogunde et al. [17]	Decision Tree (ID3)	University Crime Data	72.73%	Predict suspects from past data
Lin et al. [15]	DNN	Grid-based Crime Data	82.79%	Spatially aware DNN model
Tolan & Soliman [18]	SVM, NB, KNN	Terrorism Dataset	75.41% (SVM)	SVM best performer
Present Work	SVM+XGBoost (Hybrid)	Chicago Dataset	97.30%	High accuracy with balanced classification and prediction

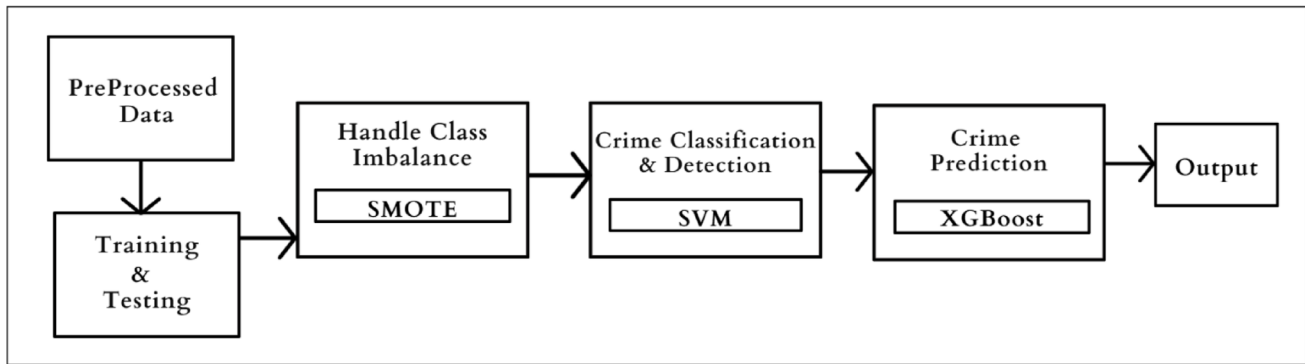


Fig. 1 Methodology Framework

crime rates. While evaluating the model's efficiency, performance metrics such as precision, accuracy, F1-score and recall are analyzed. These metrics ensures optimal performance, making a suitable system for real-world crime trend.

Dataset Description

The available Chicago Crime Dataset on Kaggle officially reported crime incidents from the Chicago Police Department system. A representative sample of 5,000 records was used. It contains detailed information on crimes recorded from 2001 to the present, excluding the most recent seven days. Each record includes attributes such as the type of crime, location description, date and time of occurrence, arrest status, and district, along with geographic coordinates to support spatial analysis. For this research, the dataset underwent preprocessing to remove duplicate entries and keep relevant attributes only. Temporal features and spatial features were worked upon to support predictive modeling. The dataset's comprehensiveness and temporal depth make it suitable for identifying all the different patterns and evaluating the performance of machine learning models in real-world.

Before model training, exploratory data analysis (EDA) was conducted on the Chicago crime dataset. Key categorical and numerical features includes offense type, location, and timestamp-based attributes. After data cleaning and feature creation, crime frequency patterns were analysed. The analysis showed that theft, assault, and narcotics-related crimes occurred most frequently, with noticeable spikes during evening hours. Correlation analysis and label encoding were applied to examine relationships among variables such as arrest status, year, and hour. The "Hour" feature emerged as a significant temporal indicator, which was later validated by XGBoost's feature importance results.

Data Preprocessing

The first step involved in preprocessing is handling the missing values. If some data points are missing, records with too many missing fields has been removed or fill in the gaps using techniques like mean, median or mode accusation. In categorical fields, missing values are filled with the most frequent category or a special "unknown" label. Thereafter, to avoid redundancy we removed duplicate records. After that, we focused on numerical variables to bring them to a uniform scale by normalization. Techniques like Min-Max Scaling or Standardization helps bring all the numeric features to a comparable scale. For categorical data, such as crime type or neighbourhood, we convert text-based values into numbers since machine learning models works better with numerical data. Finally, feature selection is performed to retain the most relevant variables. Statistical methods are used, like correlation analysis, mutual information scores, or model based techniques to identify and keep the most impactful features. Once the data is cleaned, transformed, and optimized, we split them into training and testing sets. Although the study mentions feature engineering as part of the preprocessing pipeline, it primarily focused on selecting and transforming relevant spatial and temporal attributes from the dataset rather than applying advanced dimensionality reduction techniques.

Handling Class Imbalance Using SMOTE

Here, SMOTE is applied to overcome the class imbalance problem in crime data. Crime data consists of the unbalanced distribution of crime types with some categories having fewer instances than others. The imbalance may result in crooked model predictions with the majority of classes being over emphasized but important crime types diminished. SMOTE reduces the problem by artificially creating new instances of the minority class but not just copies of existing instances. The procedure aids in developing a more balanced dataset, enhancing the model's efficiency

in identifying and classifying various types of crimes efficiently.

The oversampling ratio was set at 0.25, meaning that minority classes were synthetically expanded until they reached about a quarter of the size of the majority class. New samples were created using the k-nearest neighbors method with $k=2$, and a random seed of 42 was used to keep the results consistent across runs. This balanced yet cautious approach to oversampling helped create a more realistic dataset, allowing the classifier to learn minority class patterns without falling into overfitting. As a result, the SVM model showed improved classification performance, and XGBoost demonstrated better generalization on crime categories that were previously underrepresented.

Crime Classification & Detection Using SVM

The first stage of crime modeling uses Support Vector Machine (SVM) for crime classification. By learning patterns from historical data, SVM can distinguish between categories such as theft, assault and vandalism by finding an optimal decision boundary (hyperplane) that maximizes separation between crime classes with support vectors defining this boundary. Once the model is trained, it is tested on unseen data using metrics like accuracy, precision, recall, and F1-score to assess its generalization capability. SVM was selected because it performs well on high-dimensional data and helps reduce overfitting. Kernel functions also allow to handle certain non-linear relationships in the data. The Support Vector Machine (SVM) model was implemented using LinearSVC from scikit-learn. A linear kernel was chosen since trials with RBF and polynomial kernels offered minimal accuracy improvement but significantly increased computation time on the high-dimensional encoded data. The linear approach also improved interpretability of feature weights. The regularization parameter (C) was tuned through a small grid search over values $\{0.001, 0.005, 0.007, 0.01, 0.1\}$, with $C=0.007$ giving the best trade-off between bias and variance. The model was trained for up to 3,000 iterations to ensure convergence. This setup achieved an accuracy of 53.2% and an F1-score of 42.52%, showing that the configuration balanced performance, stability, and computational efficiency effectively.

Crime Prediction Using XGBoost

XGBoost, a powerful gradient boosting algorithm, is used to predict future crime occurrences with high accuracy. Unlike traditional regression models, XGBoost builds multiple number of trees sequentially, where each new tree rectifies the errors made by the past ones. This iterative process helps in capturing complicated. Various features are considered

such as crime type, location, time, weather conditions etc. to estimate the chances of different crimes happening in a specific area. One of XGBoost's biggest upper hand is its ability to handle large datasets efficiently while relieve overfitting through regularization techniques. This make sure that the model generalizes well with the provided data rather than memorizing patterns from historical records. A time series plot can be used for comparing the actual crime trends against the model's predictions. Time-based visualizations can also be applied to compare predicted and actual crime trends, helping reveal seasonal crime patterns or sudden spikes in criminal activity.

The XGBoost model was fine-tuned using an iterative approach with 5-fold cross-validation. A manual grid search was conducted to explore key hyper-parameters that includes max depth $\{3, 5, 7\}$, learning rate $\{0.01, 0.025, 0.05\}$, n estimators $\{10, 25, 50\}$, and the regularization terms reg_lambda and reg_alpha . The final configuration max depth=3, learning rate=0.025, n estimators=25, reg lambda=8, reg alpha=5, and subsample=0.65 offered the best balance of accuracy and stability across folds. This setup minimized overfitting and produced strong generalization.

Model Optimization and Validation

Through a combination of grid search and cross-validation hyperparameter tuning was performed. For the SVM model, parameters such as C , gamma, and kernel type were fine-tuned, with the best results obtained at $C=0.007$ using an RBF kernel. In the case of XGBoost, parameters including max_depth, learning_rate, n_estimators, lambda, and alpha were optimized using 5-fold cross-validation to ensure stability and generalization.

The dataset was split into 80% for training and 20% for testing. Model performance was evaluated using standard metrics accuracy, precision, recall, and F1-score. To make sure the reported performance metrics were not influenced by a single random train-test split, k-fold cross-validation was used during model tuning. A 5-fold cross-validation approach was used for both SVM and XGBoost during hyperparameter tuning. The dataset was split into five parts, where four folds were used for training and one for validation in each iteration. The final accuracy score was then calculated as the average across all folds, providing a more stable and reliable measure of performance. This approach helped reduce overfitting and improved the model's ability to generalize across different data segments.

Table 2 Results in default parameter value

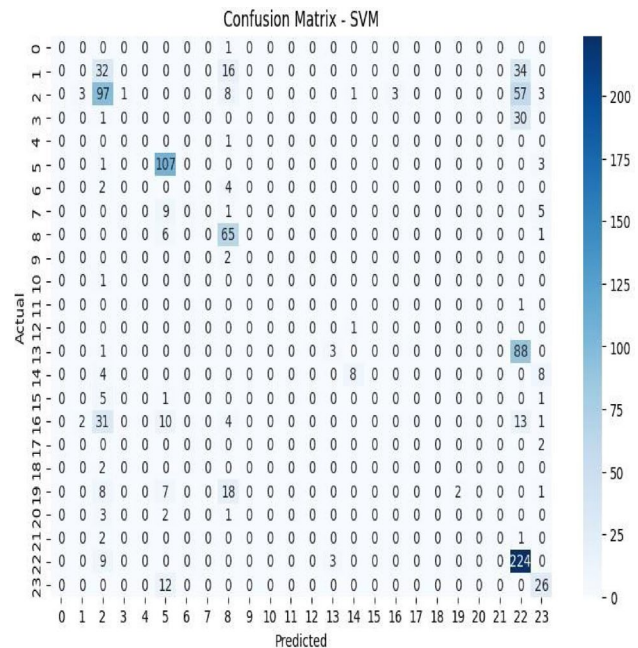
Algorithm	Accuracy	Precision	Recall	F1Score
SVM	0.532	0.48	0.53	0.4252
XGBOOST	0.973	0.98	0.97	0.9667

**Fig. 2** SVM Performance Metrics

Results and Discussion

Experimental Setup

The experiment is performed on Google Colab, using its CPU environment for running. Colab provides an easy-to-use setup with Python 3.7 and it supports major libraries like pandas, NumPy, matplotlib, scikit learn and XGBoost. The T4 instance available on Colab offers 16GB RAM, with memory usage ranging between 6GB to 8GB during run time, depending on the dataset size. The dataset used in this study comes from the publicly available “Crime Prediction in Chicago” dataset on Kaggle, which includes more than seven million reported crime records dating back to 2001. To keep the analysis computationally efficient, a representative sample of 5,000 records was randomly selected from the dataset. A fixed random seed of 42 was used to make sure that the sampling process remained consistent and reproducible across different runs. Since GPU acceleration was not used, all the processing task such as SVM and XGBoost were done on the CPU. In Google Colab, datasets can be uploaded directly using files.upload() or accessed through cloud storage for easier integration. This configuration provides an optimized and smooth workflow for both classification and prediction.

**Fig. 3** SVM Confusion Metrics**Table 3** SVM performance metrics

Algorithm	Accuracy	Precision	Recall	F1Score
SVM	53.2%	48%	53%	42.52%

The performance evaluation was done using two different machine learning algorithms, Support Vector Machine (SVM) and XGBoost. The model effectiveness was measured through metrics such as precision, accuracy, recall and F1-score. A comparative analysis, summarized in Table 2, to assess how efficient these algorithms performs with different types of dataset.

Figure 2 shows the performance analysis of support vector Machine. The SVM model was trained with a finetuned regularization parameter $C=0.007$ and a maximum iteration limit of 3000. The SVM classifier achieved an overall accuracy of 53.2%, which reflects moderate performance level. Its weighted F1-score stood at 42.52%, suggesting some imbalance between precision and recall. As shown in Fig. 3, the confusion matrix shows that while SVM correctly identified a significant share of the dataset, it still struggled to distinguish between certain crime categories. This limitation resulted in lower recall values for some classes. Precision also varied across classes, with some crimes being predicted more confidently than others. The value of recall for some classes was zero, which indicates that SVM completely failed to recognize those categories. The detailed results in Table 3 highlight both the strengths and weaknesses of SVM, pointing to areas where further optimization is possible. Since SVM performs better on linearly separable data, the complex and non-linear nature of crime

patterns suggests that approaches like kernel-based SVM or ensemble methods could be explored to improve classification accuracy

Moreover, integrating SVM with complementary models such as XGBoost, which depends on multiple decision boundaries, can help significantly improve generalization across diverse crime categories. By incorporating feature selection and leveraging hybrid methods, the system can achieve more accurate and robust classification. The XGBoost performance analysis is shown in Fig. 4. The XGBoost classifier was trained with optimized hyperparameters, including a max depth=3, learning rate=0.025, n_estimators=25, lambda (L2 regularization)=8 and alpha(L1 regularization)=5. XGBoost achieved an overall accuracy of 97.3%, significantly surpassing SVM. The weighted F1-score of 96.67% further demonstrated the model's robustness in classification. XGBoost effectively captured the intricate patterns present in the dataset. As shown in Fig. 5, the confusion matrix showed that 973 out of 1000 test samples were correctly classified, with only 27 instances misclassified. The high recall score reflected the model's ability to successfully predict actual crime occurrences while minimizing false negatives. XGBoost also maintained a high precision value, indicating prediction by the model were highly dependable. Its superior results summarized shown in Table 4, can be credited to its ensemble learning technique, which create a strong predictive model. Moreover, its capability to manage missing data and evaluate feature importance significantly improved the accuracy of classification.

A comparative analysis of the both models, illustrated in Fig. 6 clearly shows the better performance of XGBoost over SVM. While SVM faced difficulties with certain crime categories, XGBoost delivered consistently strong results across all types. Its ensemble-based approach allowed better generalization, minimized overfitting and ensured a more stable and effective classification process. The difference in the accuracy between the two models is particularly significant. XGBoost's 97.3% accuracy is a major improvement over SVM's 53.2%, making it a more reliable choice for prediction. Furthermore, the F1-score of 96.67% in XGBoost versus 42.52% in SVM highlights the advantage of ensemble learning over a single hyperplane classifier. The graphical performance comparison of the models' performance further shows XGBoost's dominance in accuracy and efficiency. The confusion matrix supports this findings, showing significantly fewer misclassifications. As summarized in Table 5, XGBoost was ultimately chosen as the final model for crime prediction. Its high precision, adaptability, and ability to generalize effectively make it a powerful and practical tool for law enforcement agencies seeking to forecast and prevent criminal activities.

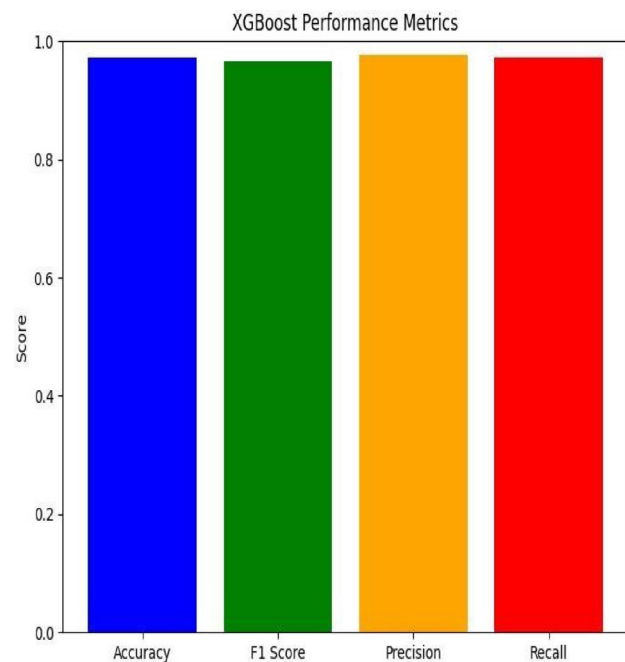


Fig. 4 XGBoost Performance Metrics

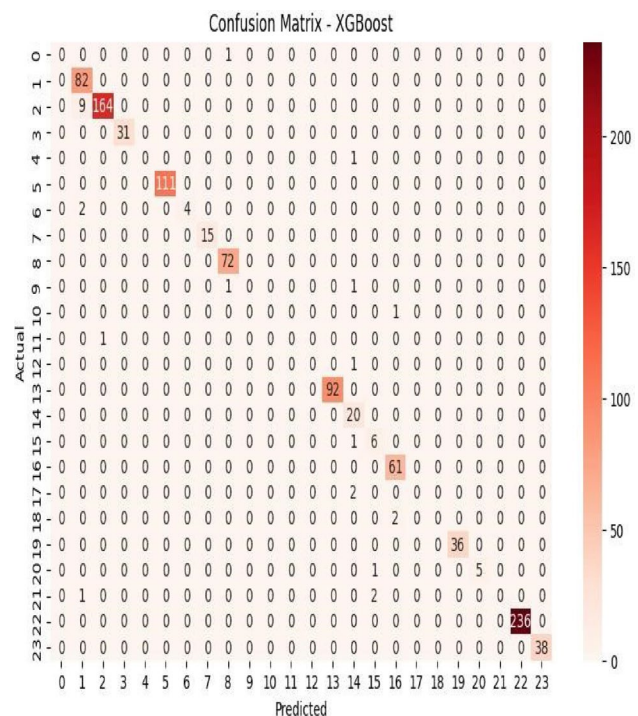


Fig. 5 XG Boost Confusion Metrics

Table 4 Performance metrics of xgboost:

Metric	Accuracy	Precision	Recall	F1-Score
XGBOOST	97.3%	98%	97%	96.67%

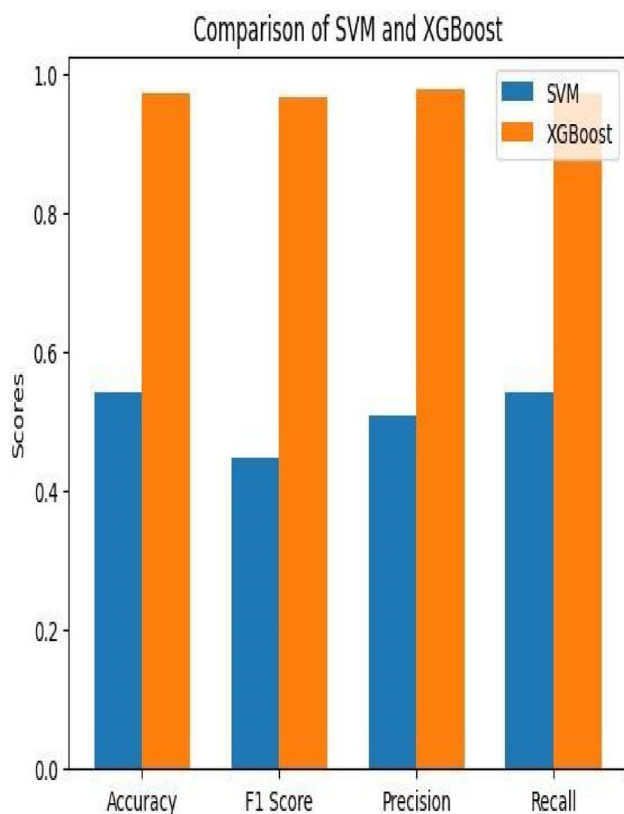


Fig. 6 SVM and XG Boost Comparison

Table 5 Comparative performance analysis

Algorithm	Accuracy	Precision	Recall	F1Score
SVM	53.2%	48%	53%	42.52%
XGBOOST	97.3%	98%	97%	96.67%

Conclusion

In conclusion, this study demonstrates the potential of machine learning techniques for crime prediction and classification to support public safety. Using historical crime data of Chicago, comparison was done between Support Vector Machine (SVM) and XGBoost for crime classification and prediction. Results indicated a wide performance gap: SVM had 53.2% accuracy with an F1-score of 42.52%. XGBoost, on the other hand, performed well with 97.3% accuracy and an F1-score of 96.67%, due to its capacity to deal with larger datasets and missing values. Offender's age, victim's age and location were the key influencing factors. Temporal elements are captured through engineered features such as Year and Hour, that help the model to identify time-based patterns, without excessive computational usage. Future research could explore time-series or deep learning models like LSTM or Temporal Convolutional Networks (TCN) to understand better about these temporal relationships and improve predictive performance.

Although formal statistical tests such as t-tests or confidence intervals were not conducted, performance differences between SVM and XGBoost were verified through multiple cross-validation folds to reduce random variation. Future work can include statistical validation using techniques like paired t-tests, Wilcoxon signed-rank tests, or confidence interval estimation to confirm the significance of these improvements and strengthen the reliability of the results. The results affirm the application of AI in predictive policing, implying law enforcement agencies can utilize such models to predict and eradicate crime successfully. While predictive policing models offer significant advancements, it is also very essential to understand their ethical, legal and privacy implications. The deployment of such systems must address transparency and fairness to withstand public trust and protect individual rights.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors have no competing interests.

Ethical Approval and Consent To Participate Not Applicable.

Consent for Publication Not applicable.

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